

Exact D.E.ⁿ

$$\frac{x^2 + y^2 + 1}{xy + x}$$

D.E.ⁿ in the form

$$\underline{M(x, y) dx} + \underline{N(x, y) dy} = 0$$

is exact if

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

and the solⁿ is

$$\int M dx + \int (\text{terms in } N \text{ free from } x) dy = C$$

or

$$\int (\text{terms in } M \text{ free from } y) dx + \int N dy = C$$

$$\frac{dy}{dx}$$

$$y = e^x$$

$$\frac{dy}{dx} = e^x$$

$$\frac{d^2y}{dx^2} - y = 0$$

$$x dy + y dx = 0$$

$$\frac{dy}{dx} = y$$

eg. $\left(\frac{d^2y}{dx^2}\right) + \left(\frac{dy}{dx}\right)^2 + y = \cos x$

❖ Solve $(e^y + 1) \cos x \, dx + e^y \sin x \, dy = 0$ $M \, dx + N \, dy = 0$

$$M = (e^y + 1) \cos x, \quad N = e^y \sin x$$

$$\frac{\partial M}{\partial y} = \cos x (e^y) \quad \frac{\partial N}{\partial x} = e^y \cos x$$

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

$\Rightarrow$ D.Eqn is exact.

Soln is $\int M \, dx + \int (\text{terms in } N \text{ free from } x) \, dy = C$

$$\Rightarrow \int (e^y + 1) \cos x \, dx + \int 0 \, dy = C$$

$$\Rightarrow (e^y + 1) \sin x = C$$

$$2(1 + x^2 \sqrt{y}) y \, dx + (x^2 \sqrt{y} + 2) x \, dy = 0$$

$$M = 2(y + x^2 y^{3/2}) \quad N = (x^2 \sqrt{y} + 2)x$$

$$\frac{\partial M}{\partial y} = 2\left(1 + x^2 \frac{3}{2} y^{1/2}\right) \quad \frac{\partial N}{\partial x} = (3x^2 \sqrt{y} + 2)$$

$$\underline{\underline{= 2\left(\frac{3}{2} x^2 y^{1/2} + 1\right)}}$$

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x} \quad \text{exact}$$

$$\int M \, dx + \int (\text{terms in } N \text{ free from } x) \, dy = C$$

$$\int 2(y + x^2 y^{3/2}) \underline{\underline{dx}} + \int 0 = C$$

$$2\left[yx + y^{3/2} \frac{x^3}{3}\right] = C$$

❖ Solve $(x^2 - x \tan^2 y + \sec^2 y) dy = (\tan y - 2xy - y) dx$

$$(\tan y - 2xy - y) dx + (x^2 - x \tan^2 y + \sec^2 y) dy = 0$$

$$M = -(\tan y - 2xy - y)$$

$$N = x^2 - x \tan^2 y + \sec^2 y$$

$$\frac{\partial M}{\partial y} = -(\sec^2 y - 2x - 1)$$

$$\frac{\partial N}{\partial x} = 2x - \tan^2 y + 0$$

$$= -\sec^2 y + 2x + 1$$

$$= 2x - (\sec^2 y - 1)$$

$$= 2x - \sec^2 y + 1$$

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

$\therefore$ D.E. is exact.

$$\int M dx + \int (\text{terms in } N \text{ free from } x) dy = C$$

$$-\int (\tan y - 2xy - y) dx + \int \sec^2 y dy = C$$

$$-\left[x \tan y - \cancel{2} y \frac{x^2}{\cancel{2}} - yx \right] + \tan y = C$$

❖ Solve $\frac{dy}{dx} = \frac{y+1}{(y+2)e^y - x}$

$$[(y+2)e^y - x] dy = (y+1) dx$$

$$(y+1) dx - [(y+2)e^y - x] dy = 0 \quad \left\{ \begin{array}{l} \int u v \\ = u \int v - \int (v u') \end{array} \right.$$

$$M = y+1 \quad N = -[(y+2)e^y - x]$$

$$\frac{\partial M}{\partial y} = 1 \quad \frac{\partial N}{\partial x} = -[-1] = 1$$

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x} \Rightarrow \text{Exact}$$

$$\int M dx + \int (\text{terms } N \text{ free from } x) dy = C$$

$$\int (y+1) dx - \int (y+2)e^y dy = C$$

$$(y+1)x - \left[(y+2)e^y - (1)e^y \right] = C$$

$$x(y+1) - [ye^y + 2e^y - e^y] = C$$

$$x(y+1) - (ye^y + e^y) = C \quad \therefore$$

$$\int u v = u \int v - \int (v u')$$

$$\int x^2 e^x$$

$$\int u v = u \int v - u' \int v_1 + u'' \int v_2$$

↳ has vanishing v_1 v_2

derivative

$$\int x^2 e^x = x^2 e^x - (2x) e^x + 2(e^x)$$

❖ Solve $e^{\frac{x}{y}} \left(1 - \frac{x}{y} \right) dy + \left(1 + e^{\frac{x}{y}} \right) dx = 0$

$$[y \sin(xy) + xy^2 \cos(xy)]dx + [x \sin(xy) + x^2y \cos(xy)]dy = 0$$

❖ Solve $[1 + \log(xy)]dx + \left[1 + \frac{x}{y}\right]dy = 0$

$$1 + \log x + \log y$$

$$M = (1 + \log(xy))$$

$$N = 1 + \frac{x}{y}$$

$$\frac{\partial M}{\partial y} = \frac{1}{xy}$$

$$\frac{\partial N}{\partial x} = \frac{1}{y}$$

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

Exact

$$\int (\text{Terms in } M \text{ free from } y) dx + \int N dy = C$$

$$\Rightarrow \int (1 + \log x) dx + \int \left(1 + \frac{x}{y}\right) dy = C$$

$$\Rightarrow x + \int (\log x)(1) dx + [y + x \log y] = C$$

$$\Rightarrow x + \int \log x(x) - \int \cancel{x} \frac{1}{\cancel{x}} dx + y + x \log y = C$$

$$\Rightarrow \cancel{x} + x \log x - \cancel{x} + y + x \log y = C$$